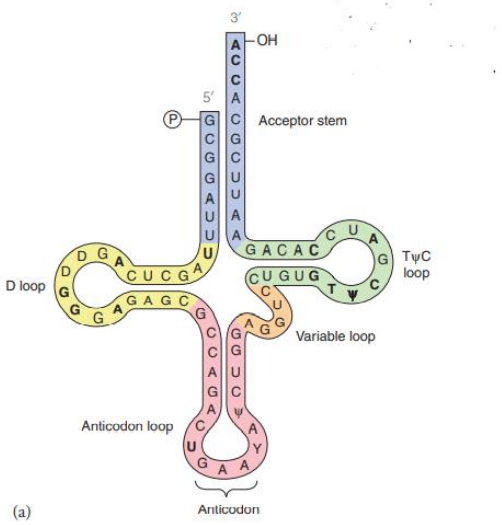
ST.PHILOMENA'S COLLEGE
B.Sc V SEM
MOLECULAR BIOLOGY
TRANSLATION
tRNA

tRNA or transfer RNA is a type of RNA, which helps in the synthesis of protein from mRNA. tRNA functions as an **adapter molecule** during the translation process. It was earlier known as soluble RNA or sRNA. As an adapter, it links the amino acids to nucleic acids. It carries the amino acid to be added in the peptide chain and also deciphers the codon for the same in the mRNA molecule.





The tRNAs are generally 76-90 nucleotides long. Each amino acid has specific tRNAs. Stop codons are not recognized by any tRNAs. The secondary structure of tRNA looks like a clover leaf and tertiary structure is like an inverted 'L' shape. The folded structure is formed due to hydrogen bonding between complementary bases.

Clover leaf structure of tRNA

The secondary folded structure of tRNA has three hairpin loops, which give it an appearance of three-leafed clover. The main constituents of tRNA are:

Acceptor arm

It is formed by the base pairing of 7-9 nucleotides of 5′ terminal and 3′ terminal. The 5′ terminal has a phosphate group and the 3′ ends with a specific sequence of CCA or CCA tail. The amino acid attaches to the 3′ hydroxyl group of the acceptor arm.

The aminoacylation of tRNA or charging of tRNA is the first step of the translation process. The enzyme aminoacyl tRNA synthetases catalyse the reaction.

DHU Loop

D arm has a stem of 3-4 base pairs and it ends in a loop called D loop as it generally contains dihydrouridine, a modified nucleotide.

Anticodon Loop

It has a 5 base pair long stem. It has an anticodon loop, which contains the complementary codon (3 nucleotides sequence) present on mRNA for the amino acid it carries. These unpaired bases of anticodon loop pair with the mRNA codon. Each codon is identified by a specific tRNA.

TΨC Loop

The T arm consists of a stem of 4-5 bp and a loop containing pseudouridine, modified uridine.

Variable Loop

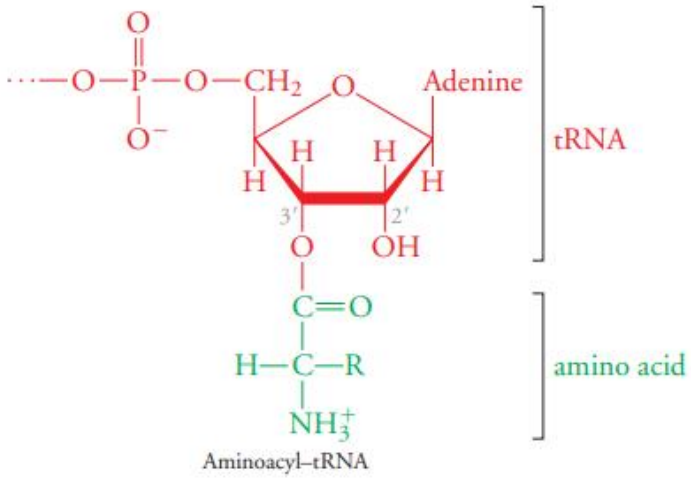
It is present between the TΨC loop and the anticodon loop. Its size varies from 3-21 bases. It helps in the recognition of the tRNA molecule.

tRNA Function

- It plays an important role in protein synthesis.
- It acts as an adapter molecule for linking amino acids to its specific codon present in mRNA.
- Aminoacylation of tRNA is the first step in protein synthesis.
- tRNA is specific to each amino acid and carries them during the translation process in the ribosomal subunits.
- The tRNA transfers the amino acid to the growing polypeptide chain in the ribosomes, which has three binding sites for tRNA, namely A, P and E, which correspond to aminoacyl, peptidyl and exit, respectively.
- This decoding of codons of mRNA by specific tRNAs continues until the entire sequence for a polypeptide chain is translated.

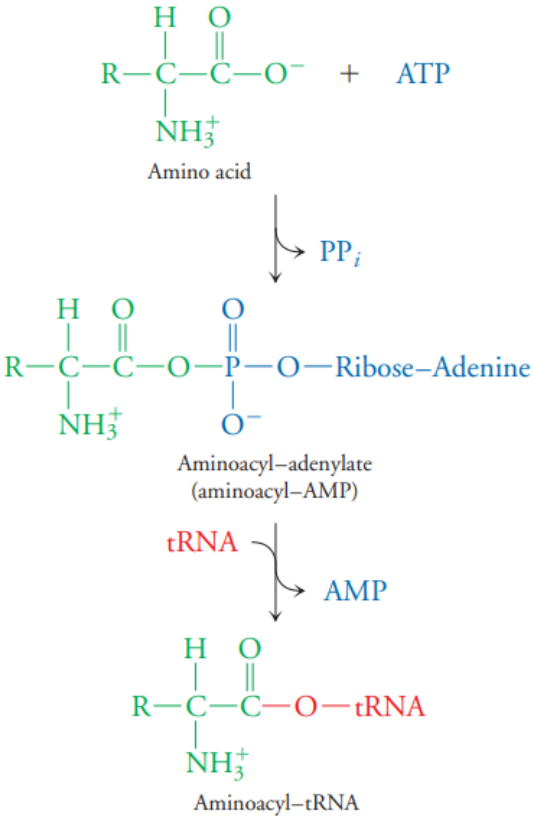
Charging of tRNA/formation of amino acyl tRNA/aminoacylation

- Aminoacylation, the attachment of an amino acid to a tRNA, is catalyzed by an aminoacyl–tRNA synthetase (AARS)
- An AARS catalyzes the formation of an ester bond between an amino acid and an OH group of the ribose at the 3′ end of a tRNA to yield an aminoacyl–tRNA.
- The tRNA molecule is then said to be “charged” with an amino acid



- The aminoacylation reaction has two steps and requires the free energy of ATP

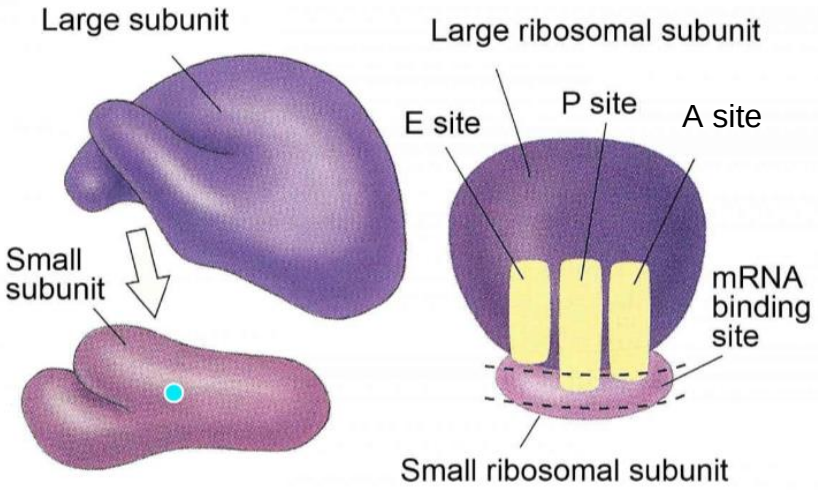
amino acid + tRNA + ATP → aminoacyl-tRNA + AMP + PPi



1. *The amino acid reacts with ATP to form an aminoacyl–adenylate (aminoacyl–AMP). The subsequent hydrolysis of the PP$_i$ product makes this step irreversible in vivo.*

2. *The amino acid, which has been "activated" by its adenylation, reacts with tRNA to form an aminoacyl–tRNA and AMP.*

Ribosome



In order to synthesize a protein, genetic information (in the form of mRNA) and amino acids (attached to tRNA) must get together so that the amino acids can be covalently linked in the specified order. This is the job of the ribosome, **a large complex containing both RNA and protein.**

	RNA	Polypeptides
***E. coli* ribosome (70S)**		
Small subunit (30S)	16S	21
Large subunit (50S)	23S, 5S	31
Mammalian ribosome (80S)		
Small subunit (40S)	18S	33
Large subunit (60S)	28S, 5.8S, 5S	49

TRANSLATION PROCESS

Like DNA replication and RNA transcription, protein synthesis can be divided into separate phases for

- Initiation,
- Elongation, and
- Termination.

These stages require an assortment of accessory proteins that bind to tRNA and to the ribosome in order to enhance the speed and accuracy of translation.

INTIATION:

- Initiation requires an initiator tRNA In both prokaryotes and eukaryotes, protein synthesis begins at an mRNA codon that specifies **methionine (AUG).**
- In prokaryotic mRNAs, this initiation codon lies about 10 bases downstream of a conserved mRNA sequence called a **Shine– Dalgarno sequence**.

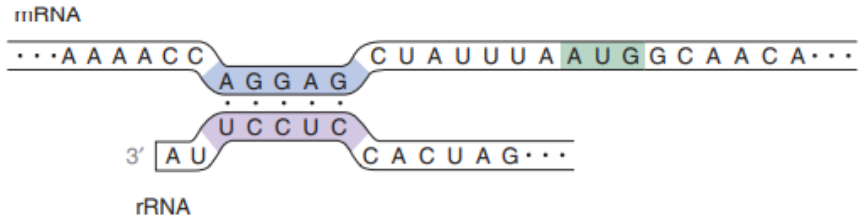
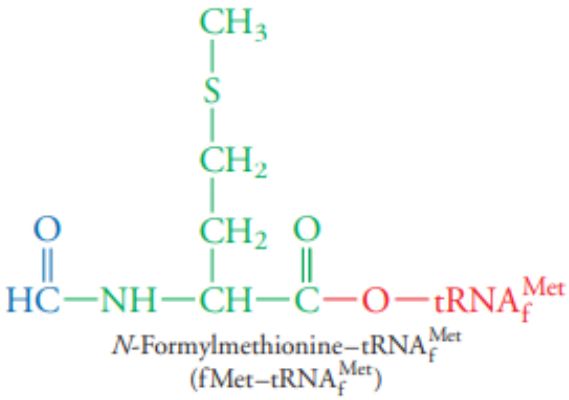
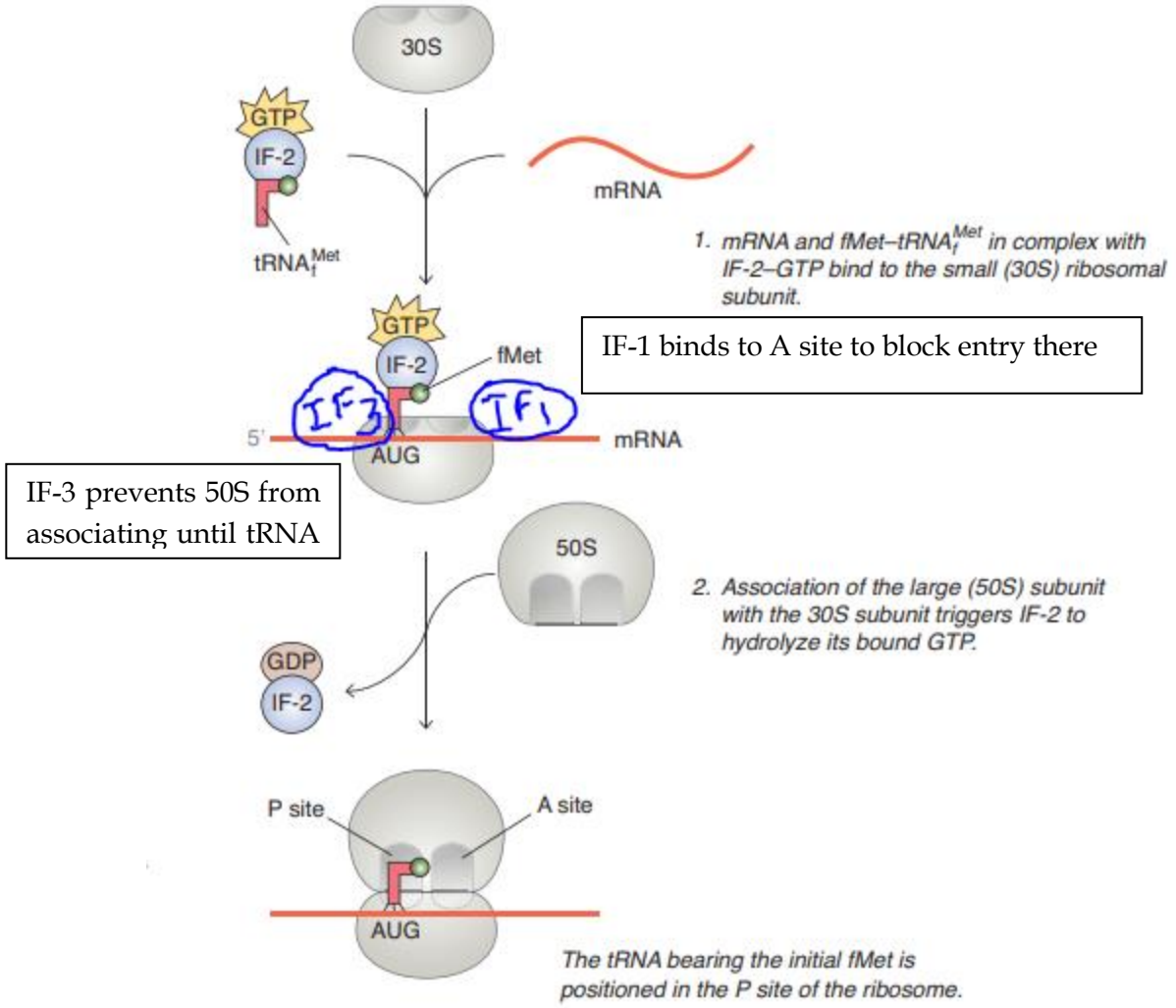


Figure 22-9 Alignment of a Shine-Dalgarno sequence with 16S rRNA. The Shine–Dalgarno sequence (blue) in an mRNA pairs with a complementary region (purple) at the 3′ end of the 16S rRNA molecule. The initiation codon is shaded green. This mRNA–rRNA interaction helps position the mRNA at the start of translation. The mRNA shown here encodes the ribosomal protein S12.

- This sequence base pairs with a complementary sequence at the 3′ end of the 16S rRNA, thereby positioning the initiation codon in the ribosome.
- Translation usually begins at the first AUG codon of an mRNA molecule.
- The initiation codon is recognized by an initiator tRNA that has been charged with methionine.
- This tRNA does not recognize other Met codons that occur elsewhere in the coding sequence of the mRNA.
- In prokaryotes, the methionine attached to the initiator tRNA is modified by the transfer of a formyl group from tetrahydrofolate . The resulting aminoacyl group is designated **f Met**, and the initiator tRNA is known as **tRNAf Met**.



NOTE: Because the amino group of f Met is derivatized, it cannot form a peptide bond. Consequently, f Met can be incorporated only at the N-terminus of a polypeptide. Later, the formyl group or the entire f Met residue may be removed.



- Initiation in prokaryotes **requires three initiation factors (IFs**) called IF-1, IF-2, and IF-3.
- IF-3 binds to the small ribosomal subunit to promote the dissociation of the large and small subunits.
- f Met–tRNAf Met binds to the 30S subunit with the assistance of IF-2, a GTP-binding protein.
- IF-1 sterically blocks the A site of the small subunit, thereby forcing the initiator tRNA into the P site.

NOTE: An mRNA molecule may bind to the 30S subunit either before or after the initiator tRNA has bound, indicating that a codon–anticodon interaction is not essential for initiating protein synthesis.

- After the 30S–mRNA–f Met–tRNAf Met complex has assembled, the 50S subunit associates with it to form the 70S ribosome.
- This change causes IF-2 to hydrolyze its bound GTP to GDP + Pi and dissociate from the ribosome.
- The ribosome is now poised—with f Met–tRNAf Met at the P site—to bind a second aminoacyl–tRNA in order to form the first peptide bond.

TABLE 22-5 Prokaryotic and Eukaryotic Translation Factors

Prokaryotic Protein	Function
IF-2	Delivers initiator tRNA to P site of ribosome
EF-Tu	Delivers aminoacyl–tRNA to A site of ribosome during elongation
EF-G	Binds to A site to promote translocation following peptide bond formation
RF-1, RF-2	Binds to A site at a stop codon and induces peptide transfer to water

ELONGATION

- In each reaction cycle of the elongation phase of protein synthesis, an aminoacyl–tRNA enters the A site of the ribosome (the initiator tRNA is the only one that enters the P site without first binding to the A site). After peptide bond formation, the tRNA moves to the P site, and then to the E site

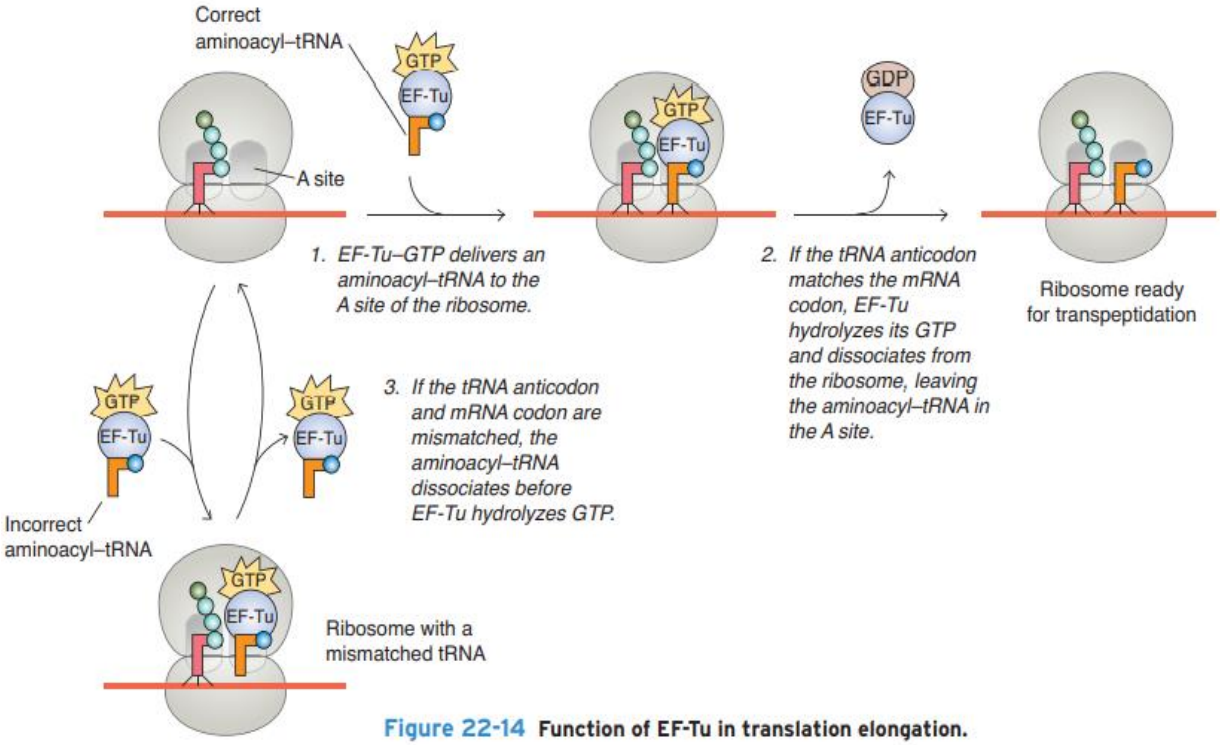
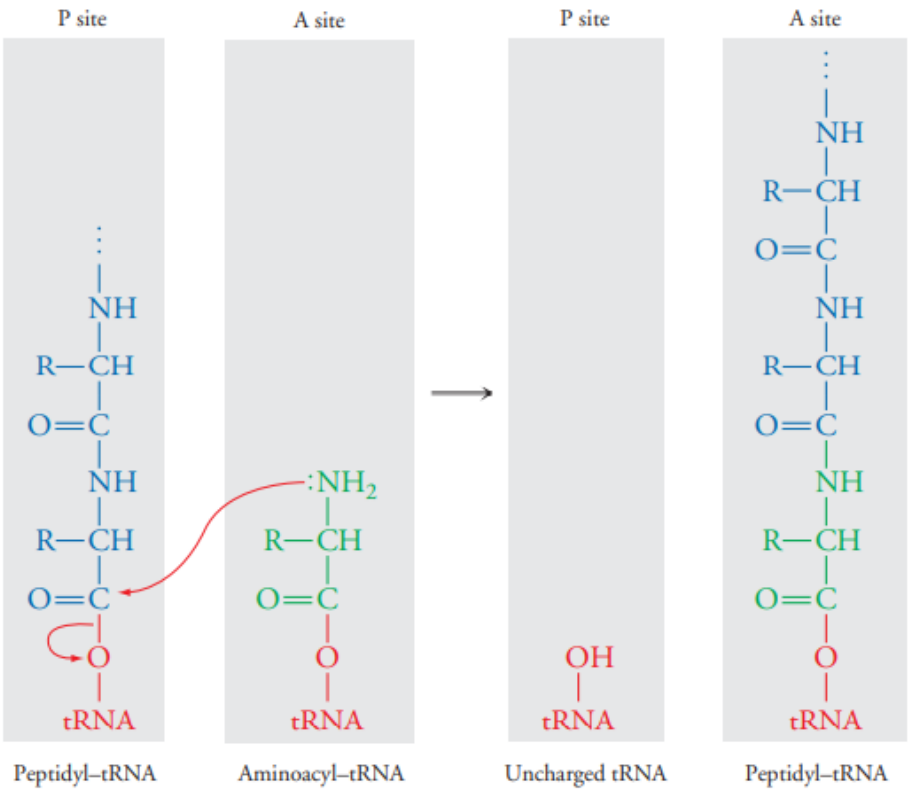


Figure 22-14 Function of EF-Tu in translation elongation.

- Aminoacyl–tRNAs are delivered to the ribosome in a complex with a GTPbinding elongation factor (EF) known as EF-Tu in E. coli
- The incoming aminoacyl–tRNA is selected on the basis of its ability to recognize a complementary mRNA codon in the A site.
- Before the 50S subunit catalyzes peptide bond formation, the ribosome must verify that the correct aminoacyl–tRNA is in place
- As the rRNA nucleotides shift to confirm a correct codon–anticodon match, the conformation of the ribosome changes in such a way that the G protein EF-Tu is induced to hydrolyze its bound GTP.
- As a result of this reaction, EF-Tu dissociates from the ribosome, leaving behind the tRNA with its aminoacyl group to be incorporated into the growing polypeptide chain
- Ribosome is now ready for transpeptidation. The peptidyl transferase active site catalyzes peptide bond formation.



- When the ribosomal A site contains an aminoacyl–tRNA and the P site contains a peptidyl–tRNA (or, prior to formation of the fi rst peptide bond, f Met–tRNAf Met), the peptidyl transferase activity of the 50S subunit catalyzes a **transpeptidation reaction** in which the free amino group of the aminoacyl–tRNA in the A site attacks the ester bond that links the peptidyl group to the tRNA in the P site
- This reaction lengthens the peptidyl group by one amino acid at its C-terminal end.
- Thus, a polypeptide grows in the N →C direction.

NOTE: No external source of free energy is required for transpeptidation because the free energy of the broken ester bond of the peptidyl–tRNA is comparable to the free energy of the newly formed peptide bond.

- During transpeptidation, the peptidyl group is transferred to the tRNA in the A site, and the P-site tRNA becomes deacylated. The new peptidyl–tRNA then moves into the P site, and the deacylated tRNA moves into the E site.
- The mRNA, which is still base paired with the peptidyl–tRNA anticodon, advances through the ribosome by one codon.
- Formation of a peptide bond causes the ribosome to loosen its grip on the mRNA.

- The movement of tRNA and mRNA, which allows the next codon to be translated, is known as translocation. This dynamic process requires the G protein called **elongation factor G (EF-G)** in E. coli.

The GTPase activity of G proteins such as EF-Tu and EF-G allows the ribosome to cycle efficiently through all the steps of translation elongation

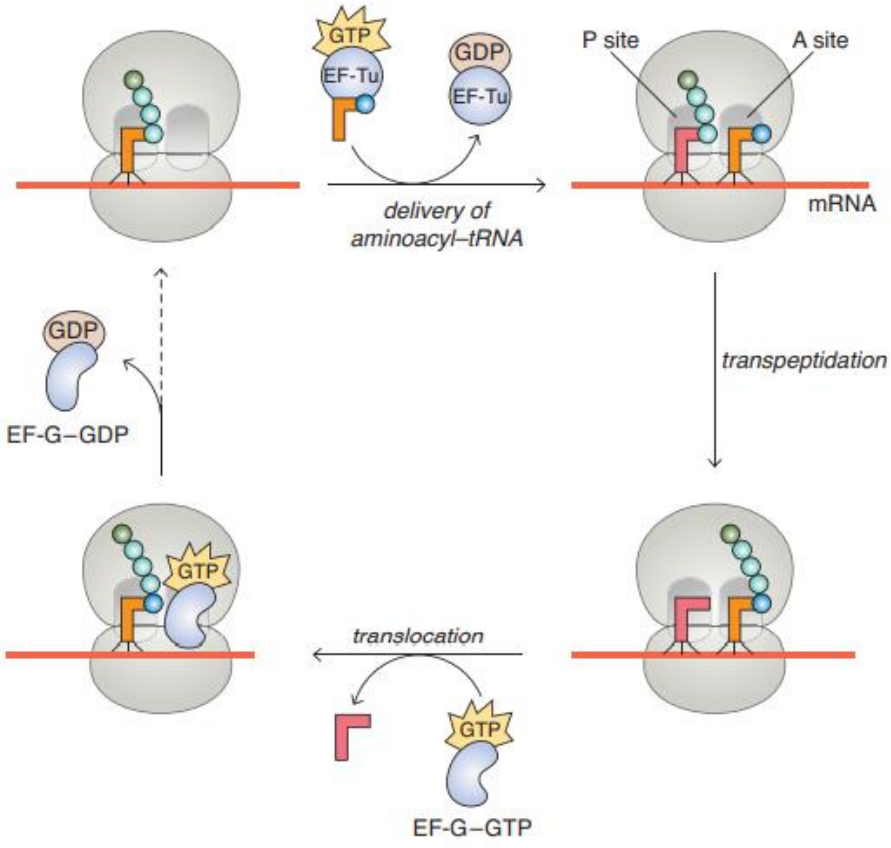
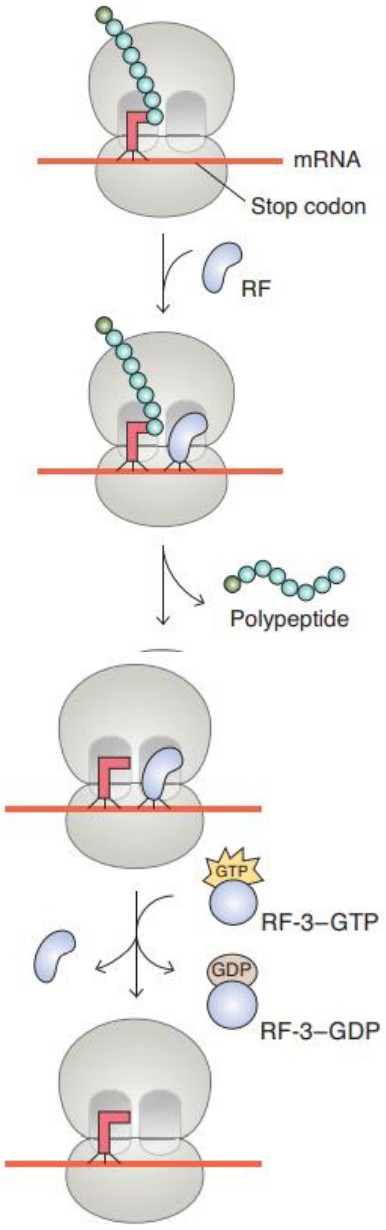


Fig: Ribosome Elongation Cycle

TERMINATION

- Translation ceases when the ribosome encounters a stop codon.
- With a stop codon in the A position, the ribosome cannot bind an aminoacyl– tRNA but instead binds a protein known as a **release factor (RF**).
  (In bacteria such as E. coli, RF-1 recognizes stop codons UAA and UAG, and RF-2 recognizes UAA and UGA.)
  - RF-1/RF-2 causes the ribosome to transfer the peptidyl group to water to release the polypeptide chain
  - GTP hydrolysis by RF-3 allows the release factors to dissociate



1. *A release factor (RF-1 or RF-2) recognizes a stop codon in the A site.*
2. *RF-1/RF-2 causes the ribosome to transfer the peptidyl group to water to release the polypeptide chain.*
3. *GTP hydrolysis by RF-3 allows the release factors to dissociate. Additional steps are required to prepare the ribosome for another round of translation.*